PROBLEM RECAP

Predict booking cancellations

based on customer characteristics and booking details

to help hotels

make smarter operational and pricing decisions

improve

profitability and guest experience

DATASET OVERVIEW

kaggle

Data Source

Hotel Booking Data
Two hotels in Portugal (City & Resort)
from July 2015 to August 2017

Data Dimensions

87,396 rows (bookings)
32 columns (features)

Target Booking Status Cancelled (1) | Not Cancelled (0)

Booking Attributes

lead time
arrival date (y/m/w/d)
days stay (wday/wend)
average daily rate
length of stay
reservation status
average daily rate
days in waitlist
deposit type
reservation status
reservation status date
distribution channel
agent
company

Customer Attributes

# of adults
# of children
# of babies
country of origin
customer type
market segment

Hotel/Room Details

hotel type
meal preference
reserved vs assigned
# of parking
# of special request

Booking Behavior

previous cancellation
repeated guest
booking changes

CLASSIFICATION MODELS

Predict if booking is cancelled or not

- Random Forest
- Gradient Boosting
- K-Nearest Neighbor (KNN)
- Support Vector Model
- Logistic Regression

RANDOM FOREST

Predicts based on multiple booking cancellation flowcharts (trees) created

- Works well with numerical and categorical (e.g., ADR, lead_time, country, deposit type)
- Captures **non-linear** booking behavior (interactions)
  - **Linear:** if Lead time ↑, cancellation likelihood ↑
  - **Non Linear (Interactions)**
    - Lead Time ↑ + no deposit → High Cancel
    - Lead Time ↓ + deposit paid → Low Cancel
- Handles many **variables without degradation**
  - Naturally filters "useless" or "noisy" variables
- Can identify **top drivers** for booking → Relevant to analytics

MODEL WEAKNESS

- **Overfitting Risk**

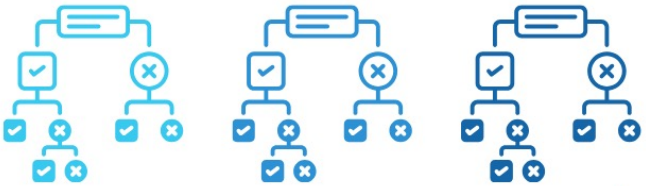
- **Harder to interpret**
  - Can't generate a Single Tree Visualization **vs Decision Tree**
  - Cannot trace the exact reasoning behind each prediction (black box) vs **Logit Regression**

GRADIENT BOOSTING

Builds trees, and **new trees fix errors** of previous trees.

Hotel Data is **full of exceptions** (corrections)

- Long lead time = cancel
- Exception: Unless repeated guests



Improves as it corrects

Predicts Well as it learns from corrections

Can provide **top drivers in cancellation**

MODEL WEAKNESS

- Overfitting
- Sensitive to Outliers
- **Hard to tune**
- Long computing time
- Additional Feature Engineering

K-NEAREST NEIGHBOOR

Booking history repeats itself

Somewhat close to who cancelled previously → cance

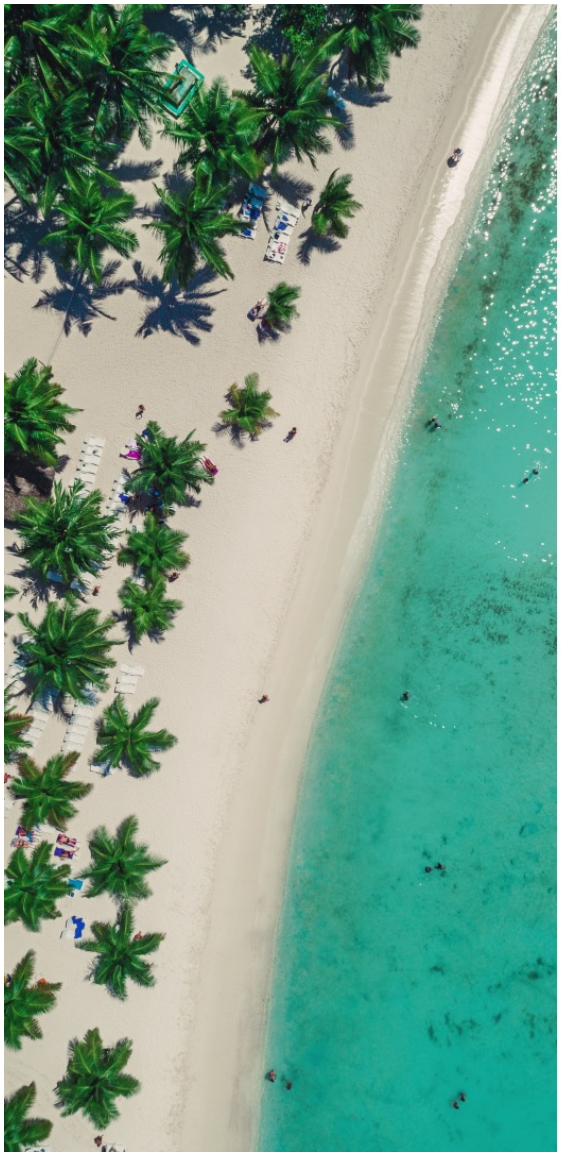
- **Fits the scenario**: Guests often fall into "clusters"

- Mimics **Customer Persona** through Look-alikes
- Can capture **interactions**

MODEL WEAKNESS

- **No Models**: Lazy learner
  - It does **capture** cancellations
  - BUT **not the relationship** between factors and cancellation

- **Dimensionality** Curse (32)
- **Equality flaw:** Treat each similarly



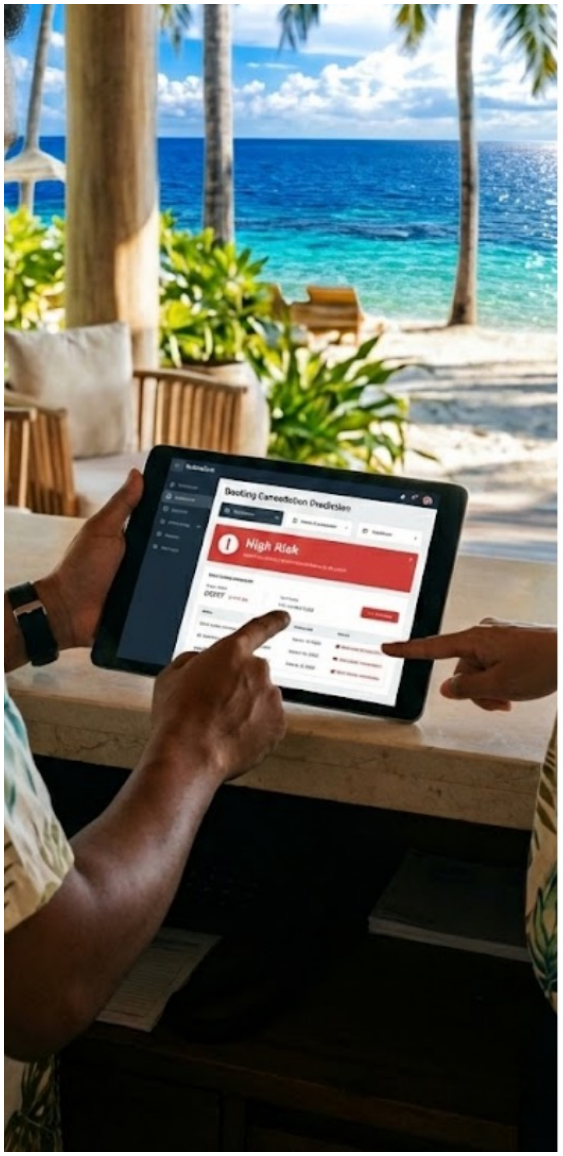
SUPPORT VECTOR MACHINES (SVM)

Classifies using best boundary (widest margin) between cancelled booking & not

- Works well with numerical and categorical
- **Robust to overfitting**
  - Focuses on **CONFUSING GUESTS** (boundary) which are hardest to predict
    - Lead time=60
    - Deposit: Refundable
  - Vs Logit: **Doesnt Flip Randomly**
    - if booked 61 days
- Captures **non-linear pattern** (interactions)
  - High Lead Time+ No Deposit → Cancellations
- **Resistant to Outliers**
  - Since it focuses on boundary
- **Can handle imbalance effectively**
  - 27% Cancellation

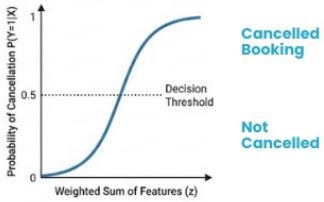
MODEL WEAKNESS

- Doesn't Predict **Likelihood**
- **Lack** of native feature **importance**
- **Unusual bookings** (that does not follow normal patterns) prone to misclassification



LOGISTIC REGRESSION

- Clear **Binary** (Cancelled vs Not) Outcome



- Works well with:
  - **Numeric**: ADR, Lead Time
  - **Categorical**: Repeat? Hotel Type
- Provides **Odds/ Likelihood** every unit

How much **lead time** increases the cancellation odds?

Are domestic guests more likely to cancel? How much more likely?

- Can handle large data n=87k efficiently

MODEL WEAKNESS

- Doesn't automatically detect **interactions**

Lead Time

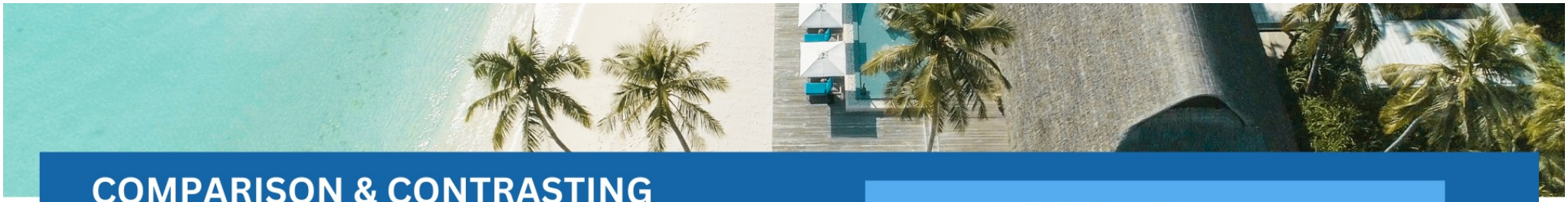
Special Request

- Vulnerable to **multicollinearity** and **outliers**

Market Segment

Dist Channel

- Rigid probability threshold (0.5)
  - (e.g., flag only high-risk cancellations).



COMPARISON & CONTRASTING

SUMMARY

Criterion	Logistic Regression	Gradient Boosting	Random Forest	SVM	KNN
Interpretability	🟢 High	🔴 Low	🔴 Low	🔴 Low	🟡 Medium
Provides a Model?	🟢 Yes	🟢 Yes	🟢 Yes	🟢 Yes	🔴 No
Feature Importance	🟢 Yes	🟢 Yes	🟢 Yes	🔴 No	🔴 No
Non-Linear Capability	🔴 Poor	🟢 Excellent	🟢 Excellent	🟢 Excellent	🟡 Good
High-Dimensional Performance	🟡 Good	🟢 Excellent	🟢 Excellent	🟢 Excellent	🔴 Poor
Noise Handling	🟡 Medium	🟢 Good	🟢 Excellent	🔴 Poor	🔴 Very Poor
Handles Imbalance	🟢 Yes	🟢 Yes	🟢 Yes	🟢 Yes	🔴 Poor
Overall Ranking	**1st**	**2nd**	**3rd**	**4th**	**5th**

Logistic Regression is our primary model because its interpretability allows us to explain how each booking characteristic influences cancellation risk.

insightout

Decoding Data from Insights to Impact

ADVANTAGES & DISADVANATGES

SUMMARY & CHOICE

	Logistic Regression	Gradient Boosting	Random Forest	SVM	KNN
Advantages	Explains how features work & generates likelihood	High Accuracy. Finds weak patterns	Detects Non Linear Robust	Good on Interactons and Outliers	Captures Clusters (Non Linear)
Disadvantages	Missess Interactions Sensitive to Outliers	Difficult to tune. Outlier Sensitive	Challenging to Explain	No Feature Importance	Poor in Many Features particularly dimensions
Reason for Usage	Primary model for explainability	Primary choice for maximum predictive accuracy	Identifying Variables and Robustness	Classifying mid-looking instances	If there are clusters in the data and is captured